SAPIENZA
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Intelligenza Artificiale

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Esercitazione A.3.3 (E.A.3.3)

Parte A

Esplorazione di Spazi degli Stati

HP 2D-Protein Folding

– Solo Testo –

1 Objectives

The *HP 2D-Protein Folding* problem models a simplified version of a well-known problem in computational biology. It consists in finding the spatial conformation of a protein (*i.e.*, a sequence of amino-acids) with minimal energy.

The simplifications with respect to the real problem are twofold.

First, the 20-letter alphabet of amino-acids is reduced to a two-letter alphabet, namely H and P. H represents *hydrophobic* amino-acids, whereas P represents polar or *hydrophilic* amino-acids.

Second, the conformation of the protein is limited to a bi-dimensional discrete space.

Nonetheless, these limitations have been proven to be very useful for attacking the whole protein conformation prediction problem. The simplified version is known to be NP-complete [**?**] and very hard to solve in practice.

Given the sequence of amino-acids of the protein, *i.e.*, a string over {H,P} of length n, the problem aims to find a connected shape for it on a 2D grid (with coordinates in $[-(n-1),(n-1)]$, starting at a pre-determined position, *e.g.*, $(0,0)$), which is non-crossing, and such that the number of "contacts", *i.e.*, the number of non-sequential pairs of H's for which the Euclidean distance of the positions is 1 is maximized (the overall energy is the opposite of the number of contacts). Figure 1.1 shows a possible conformation of the protein "PHHPHPPHP", with overall energy -2.

The goal of this exercise is to define pseudo-code (and possibly an implementation in any high-level language) of a portfolio of programs that, given an instance of the problem (namely, a value for positive integer $n$and the linear structure of a protein in the form of a string over symbols 'H' and 'P') find an optimal 2D conformation of the input protein. Each program in the portfolio shall use a solving method inspired by a non-informed/informed search state exploration algorithm shown in the course, and in particular:

1. depth first search

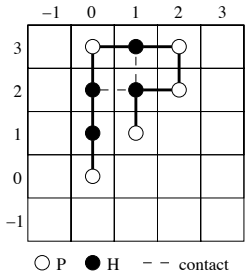

Figure 1.1: HP 2D-Protein folding problem: A possible conformation for protein "PHHPHPPHP" with two contacts, and overall energy -2.

2. breadth first search
3. min cost search
4. iterative deepening search
5. best-first greedy search
6. A*.

In case of informed search algorithms, a suitable heuristic function shall be designed.